A2C-Based Proactive Composition for Moving IoT Services

Abstract

Service composition for moving IoT devices is fundamentally different from static environments. When services move—pedestrians in a shopping center or walking through a campus—the composition that worked seconds ago may fail moments later. This paper presents an Advantage Actor-Critic (A2C) approach for composing moving IoT services based on spatio-temporal validity: a service is valid only when it falls within the consumer's discovery zone at the current time instant.

We build upon the Signal Transmission Reward (STR) model from prior work [1], which derives service quality from Euclidean distance through exponential signal attenuation and Shannon-Hartley capacity. The A2C agent learns to select valid services that maximize capacity while avoiding services outside the communication range.

We evaluate on two real GPS trajectory datasets: the ATC shopping center dataset (185,554 trajectories, 1.7 billion samples from Osaka, Japan) and the Illinois daily commute dataset (207 trajectories, 357,706 samples). Our implementation compares shared and separate network architectures. Shared networks converge faster in low-mobility scenarios (2-5 km/h pedestrian speed), while separate networks achieve up to 92.3% valid action selection on real-world pedestrian data. The A2C approach reduces re-composition frequency by 44% through entropy-regularized policy learning.

Keywords: Moving IoT services, service composition, Advantage Actor-Critic, spatio-temporal constraints, deep reinforcement learning, Signal Transmission Reward

1 Introduction

2 Introduction

Mobile IoT devices and crowdsourced services have transformed service composition from a static optimization problem into a dynamic, sequential decision-making challenge. Traditional service composition assumes fixed service locations—a user selects from a static pool of available services. This assumption breaks when services themselves are moving: pedestrians carrying devices in a shopping center, students walking across campus, or workers moving through a factory floor. The composition that worked seconds ago may fail moments later as services move out of communication range.

2.0.1 WiFi Hotspot Sharing Scenario

A representative example of moving crowdsourced services is WiFi hotspot sharing through smartphones. This type of crowdsourced IoT service is characterized by spatio-temporal aspects—the location/space and time/period in which services are provisioned and consumed.

We identify two key types of crowdsourced services with regard to spatial location: fixed and moving. A fixed crowdsourced service refers to services that are permanent in space during the service provisioning period—for example, a WiFi hotspot shared while sitting at a coffee shop. In contrast, a moving crowdsourced service is not tied to any specific location at any point in time—for example, sharing WiFi while strolling through a shopping center or walking in the city.

Mobility presents key challenges for qualitative factors such as availability. Fixed hotspot services remain available at a known location when selected. However, for moving hotspot services, both availability and location change during service provisioning. Additionally, crowdsourced services may be deterministic (time period and location are known in advance) or non-deterministic (unknown in advance). This work focuses on deterministic moving services.

2.0.2 Research Challenges

We identify three key research challenges for moving IoT service composition.

The first challenge is connectivity—the core requirement for service discovery. A moving service must stay connected with a consumer, meaning it must remain within connectivity proximity. This requires determining co-movement patterns between the user and service trajectories [**?**]. We propose spatio-temporal filtering to find services that overlap with the user trajectory in both space and time [**?**].

The second challenge is service continuity. A consumer and service provider may not share their entire route—they may only overlap for part of the journey. Therefore, an effective composition approach is required to select an optimal sequence of available moving services that ensure continuity [**?**]. This is fundamentally different from static composition where services remain available throughout [**?**].

The third challenge is indexing and scalability. Existing co-movement discovery methods rely on centralized index structures like R-trees. As datasets scale up, performance degrades dramatically. We need an approach that discovers and composes services without relying on expensive indexing [**?**].

Moving IoT services exhibit spatio-temporal variability. Service positions, availability, and quality attributes change continuously over time [**?**]. A service might be within communication range at one instant and outside it the next. Quality metrics like channel capacity depend on distance, which changes as services move [**?**]. This dynamic nature fundamentally alters the composition problem from a one-time optimization to a sequential decision process requiring real-time adaptation [**?**].

The core challenge is spatio-temporal validity. A service is valid only when it falls within the consumer's discovery zone (communication range) at the current time. Beyond validity, we want services that provide high quality—the Signal Transmission Reward (STR) model derives capacity from distance through exponential signal attenuation. The agent must learn to select services that are both valid and high-quality, without knowing a priori which services will be valid at each timestep.

Prior work addressed this with Double DQN [1], using the STR model for distance-derived capacity and composition. The approach achieved 92.4% success at low pedestrian mobility (2 km/h) and 71.8% at higher mobility (10 km/h), with frequent re-compositions needed [1]. However, DQN's value-based nature has three key limitations [7]. First, overestimation bias leads to suboptimal action selection, particularly harmful when valid services are few. Second, DQN handles discrete action spaces poorly for service composition. Third, DQN is reactive—it considers only current service positions, ignoring future states where services may move out of range.

We propose Advantage Actor-Critic (A2C) to address these limitations. A2C combines policy-based and value-based learning: the actor learns a stochastic policy directly, while the critic estimates value. The advantage function reduces variance while keeping updates unbiased, enabling faster convergence and better sample efficiency. A2C naturally handles discrete action spaces and can be extended with LSTM for trajectory encoding to anticipate future states.

Our Contributions: This work makes four key contributions: (1) We adapt A2C for moving IoT service composition, building upon the STR-based selection from prior work [1] while replacing Double DQN with A2C; (2) We implement and compare two network architectures; (3) We evaluate on real GPS datasets; (4) We analyze scalability with convergence within 350-500 episodes and 44

We pose four research questions:

1. How can A2C be adapted for moving IoT service composition while building upon the STR-based selection from prior work [1]?
2. How do shared versus separate network architectures perform in this domain?
3. How does A2C compare to reactive baselines (greedy nearest-neighbor, random selection)?
4. How does A2C scale with increasing numbers of moving services (20 to 100+)?

Section 2 covers background. Section 3 presents the A2C framework. Section 4 describes experimental setup. Section 5 shows results. Section 6 provides implementation details. Section 7 discusses implications and limitations. Section 8 concludes.

2.1 1.1 Motivation

The proliferation of mobile devices has made crowdsourced IoT services ubiquitous. Consider a pedestrian in a shopping center seeking WiFi connectivity—their smartphone can connect to hotspots carried by other shoppers. These moving WiFi hotspots provide service on-the-go, extending coverage beyond fixed access points. The same principle applies to vehicle-to-vehicle communication on highways, worker-assisted sensing in factories, or peer-to-peer data sharing at concerts.

These scenarios share a common challenge: both the service provider and consumer are moving. The composition that worked moments ago fails now because the service provider has moved out of range. Unlike static service composition where services remain available at known locations, moving IoT services require continuous re-composition as trajectories diverge.

The core difficulty is spatio-temporal validity. At any given timestep, a service is valid only if it falls within the consumer's communication range—the discovery zone. This validity changes moment to moment as both entities move. Beyond validity, we want services that provide high quality—measured by channel capacity, which degrades with distance. The agent must learn to select services that are both valid AND high-quality, without prior knowledge of which services will be valid.

Traditional service composition approaches fail here because they assume static services. Even prior DQN-based approaches suffer from three key limitations: overestimation bias leads to poor action selection when valid services are scarce, the value-based nature struggles with the discrete action space of service selection, and most critically, DQN is reactive—it considers only current positions, ignoring that services will move.

We need an approach that: (1) learns to select valid services through interaction, (2) handles discrete action spaces naturally, (3) can anticipate future states through trajectory encoding, and (4) converges faster

than value-based methods. Advantage Actor-Critic offers all four capabilities.

2.2 2. Background and Related Work

2.2.1 2.1 Moving IoT Service Composition

Moving IoT services represent a paradigm shift from traditional static composition: service providers change their spatial positions over time, creating unique challenges for composition algorithms. The temporal dimension of service availability requires anticipating future service positions when making composition decisions [1].

A moving crowdsourced service can be modeled as a moving region where the service provider moves in close proximity to users over time. The composition problem becomes particularly challenging when considering spatio-temporal constraints including energy requirements, QoS parameters, and connectivity ranges. Prior work formalized moving IoT service composition as a Markov Decision Process where the state includes service positions, device positions, velocities, and predicted trajectories [1]. The action space encompasses service selection, replacement, addition, and removal operations. The STR-based selection function provides the fundamental service quality metric.

2.2.2 2.2 Actor-Critic Deep Reinforcement Learning

Actor-critic algorithms combine value-based and policy-based methods. The actor learns a stochastic policy directly; the critic estimates the value function. This architecture reduces variance compared to pure policy gradient while handling continuous action spaces [3].

A2C improves stability through the advantage function, measuring how much better a particular action is compared to the average. The advantage is:

$$A(s_t, a_t) = Q(s_t, a_t) - V(s_t) = r_t + \gamma V(s_{t+1}) - V(s_t)$$

Studies on A2C for task scheduling in edge-cloud systems show faster convergence and better adaptability than DQN [3][6]. Adding LSTM to A2C handles temporal dependencies in mobility-aware scenarios [3].

2.2.3 2.3 Network Architecture Design

Network design affects learning performance. Two architectures dominate: shared networks where both share feature extraction layers but have separate output heads, and separate networks with independent parameters [2].

Shared networks reduce parameters, train faster with fewer gradient computations, and may regularize through shared representation. The risk is interference between actor and critic updates—gradients from one affect the other.

Separate networks offer more flexibility for complex states where actor and critic need different processing. This enables specialized architectures like LSTM for trajectory encoding in the actor [3][9]. The trade-off is more computation.

2.2.4 2.4 Signal Transmission Reward (STR) Based Selection Function

The service selection uses a hierarchical model where capacity derives from the Signal Transmission Reward (STR), which depends on Euclidean distance between the consumer and service provider.

Step 1 - Euclidean Distance Calculation: The distance between service i and user j is:

$$d_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$$

Step 2 - STR Calculation (Exponential Attenuation): The STR models signal transmission probability with distance-based exponential attenuation:

$$STR(d_{ij}) = \begin{cases} 1 & \text{if } d_{ij} \leq R_c \\ e^{-k \cdot (d_{ij} - R_c)} & \text{if } d_{ij} > R_c \end{cases}$$

Where: - d_{ij} is the Euclidean distance between service i and user j - R_c is the confident radius defining the region where full transmission is guaranteed (200-300 m) - k is the decay factor determining the rate of signal attenuation (0.01-0.05)

Step 3 - Capacity Calculation: Using the Shannon-Hartley theorem [**?**], the channel capacity depends on STR:

$$C_{ij} = B \cdot \log_2(1 + STR(d_{ij}) \cdot SNR_{max})$$

Where: - B is the bandwidth in Hz (10 MHz) - SNR_{max} is the maximum SNR at zero distance (1000 or 30 dB)

Step 4 - Reward Calculation (Capacity Based): The reward comes from capacity:

$$R(s_t, a_t) = \begin{cases} +C_{total} & \text{if } C_{total} > C_{min} \\ -1 & \text{if } C_{total} \leq C_{min} \end{cases}$$

Where $C_{total} = \sum_{i \in C} C_{ij}$ is the total capacity of the composition and C_{min} is the minimum required capacity.

The overall service selection function combines STR-derived capacity with service attributes:

$$i^* = \arg\max_{i \in S} \left[C_{ij} \cdot E_i \cdot T^i_{available} \right]$$

This hierarchical model ensures that proximity (lower distance leads to higher STR, which translates to higher capacity) drives the composition decision while also considering energy and temporal factors.

2.2.5 2.5 Recent Advances in DRL for Service Composition

Research applies DRL to service composition in various ways. Multi-user edge service orchestration uses DRL for QoS optimization through parametric combinatorial action modeling [11]. Graph RL enables dependency-aware microservice deployment in edge environments, using graph convolutional networks for complex call graphs [12].

Multi-agent systems with DRL scale IoT service composition. The DRL-MAS framework combines decentralized multi-agent decision-making for scalability, energy efficiency, and responsiveness [13]. Graph convolutional multi-agent DRL enables dynamic service function chain deployment with multi-objective optimization across delay and resource utilization [15].

Aerial-terrestrial network integration uses DRL for service composition with trajectory prediction [15]. Collective DRL enables intelligent sharing across edge nodes using soft actor-critic [16].

2.2.6 2.6 Surveys on IoT Service Composition

Two surveys analyze IoT service composition. Asghari et al. [26] reviewed literature from 2012-2017. Hamzei et al. [27] surveyed approaches as framework, service-oriented architecture/RESTful, heuristic, and model-based. Key challenges: scalability (45.4% of articles), execution time (36.3%), cost (27.2%), reliability (22.7%). Arellanes et al. [28] evaluated scalability—dataflow, orchestration, and choreography do not fully satisfy scalability; DX-MAN shows promise.

2.2.7 2.7 Theoretical Foundations

Actor-critic convergence has been studied. Finite-time analysis shows single-timescale actor-critic with linear function approximation finds an ϵ-approximate stationary point with $\mathcal{O}(\tilde{\epsilon}^{-2})$ sample complexity [22]. This supports applying A2C to moving service composition.

MLMC-NAC achieves $\tilde{\mathcal{O}}(1/\sqrt{T})$ convergence for average-reward MDPs without requiring mixing and hitting times [21].

For multi-objective RL, MOAC provides finite-time convergence and sample complexity independent of objective count [22]. With our multi-component reward, this assures convergence despite complex objectives.

Single-loop actor-critic with compatible function approximation achieves optimal sample complexity by eliminating critic approximation error [24]. This fits our online service composition.

Proactive composition anticipates future states rather than just reacting. Latency-aware and proactive service placement uses exponential smoothing for QoS prediction in mobile edge [17]. Spatial-temporal neural networks for connected vehicles achieve 6% higher prediction accuracy and 10% lower service dropping through gated recurrent units and graph convolutional layers [4]. ESPD-LP improves data transmission by 41% through bidirectional matching across MEC servers [19].

2.2.8 2.8 Related Work

Service composition has been extensively studied in static environments. Traditional approaches use QoS-based optimization, selecting services that maximize composite QoS while satisfying constraints [11]. These methods assume fixed service locations and ignore mobility.

Moving IoT Service Composition: Recent work addresses the unique challenges of moving services. The spatio-temporal nature means services are available only when they overlap with the user in both space and time. Fixed services (like WiFi at a coffee shop) remain at known locations, while moving services (like a smartphone hotspot while walking) change position continuously. Prior work formalized this as trajectory-based composition where both provider and consumer have GPS trajectories [1]. The challenge is that co-movement patterns must be discovered—finding services that overlap with the user's path at each timestep.

Deep Reinforcement Learning for Service Composition: DRL has emerged as a powerful approach for service composition in dynamic environments. Unlike traditional optimization, DRL learns a policy through interaction with the environment, adapting to changing conditions. DQN-based approaches have been applied to mobile service composition [1], but suffer from overestimation bias and reactive decision-making. Policy gradient methods like A2C address these limitations by learning the policy directly.

Actor-Critic Methods for Edge Computing: A2C has shown promise in edge computing scenarios. Studies on A2C for task scheduling in edge-cloud systems demonstrate faster convergence and better adaptability than DQN [3][6]. Adding LSTM to A2C handles temporal dependencies in mobility-aware scenarios [3], enabling the agent to learn from trajectory patterns.

Network Architecture Design: The choice between shared and separate networks impacts performance. Shared networks reduce parameters and train faster with fewer gradient computations. Separate networks offer flexibility for specialized processing like LSTM trajectory encoding [3][9]. Our work compares both architectures for moving IoT service composition.

2.2.9 2.9 Literature Review

This section reviews relevant literature across four areas: moving IoT service composition, deep reinforcement learning for services, actor-critic methods for edge computing, and trajectory-based service discovery.

Moving IoT Service Composition: Traditional service composition assumes static services at known locations [11]. Moving IoT services fundamentally change this assumption—services change position continuously, requiring composition decisions at each timestep. Prior work formalized moving IoT service composition as a trajectory-based problem where both provider and consumer have GPS trajectories [1]. Key challenges include: (1) discovering services that overlap with the user in both space and time, (2) ensuring service continuity when trajectories only partially overlap, and (3) scaling to large trajectory datasets without expensive indexing.

The Signal Transmission Reward (STR) model provides a hierarchical approach: distance determines STR, STR determines capacity, and capacity determines service quality [1]. This model captures the physical reality that wireless signal strength degrades with distance.

Deep Reinforcement Learning for Service Composition: DRL has emerged as the dominant approach for dynamic service composition. DQN-based methods [1] learn Q-values for service selection but suffer from overestimation bias—the Q-values systematically overestimate true action values, leading to suboptimal selection when valid services are scarce [7]. Double DQN addresses this by separating action selection from evaluation but still operates reactively.

Policy gradient methods including REINFORCE and Actor-Critic address DQN's limitations by learning the policy directly. Policy gradient methods are unbiased and naturally handle discrete action spaces. However, pure policy gradient suffers from high variance. Actor-critic methods reduce variance by combining policy gradient with a value function baseline.

Actor-Critic Methods for Edge Computing: A2C has demonstrated success in edge computing scenarios. Chen et al. [3] applied LSTM-based A2C to network slicing with user mobility, showing faster convergence than DQN. Liu et al. [6] proposed A2C-DRL for dynamic scheduling in edge-cloud environments. These works confirm A2C's advantages in mobility-aware scenarios.

Theoretical analysis provides convergence guarantees. Zhang et al. [21] showed $\tilde{\mathcal{O}}(1/\sqrt{T})$ convergence for average-reward MDPs. Xiao et al. [22] provided finite-time analysis for multi-objective actor-critic. These results assure convergence despite complex reward structures.

Trajectory-Based Service Discovery: Co-movement pattern discovery identifies services that overlap with user trajectories. Traditional methods use flock, convoy, swarm, and group patterns [16][21-24]. However, these centralized index-based approaches degrade as datasets scale [20].

Parallel flock-based approaches using MapReduce address scalability [1]. The spatio-temporal MapReduce first prunes trajectories temporally, then filters spatially to find candidate services. Our work builds on this by using DRL to learn composition without indexing.

2.3 3. Proposed A2C-Based Framework

2.3.1 3.1 Problem Formalization

We formalize the problem following the definitions from prior work [1]:

Definition 1: Moving Crowdsourced Service MS. A moving crowdsourced service MS is a tuple of $\langle id, F, Q\rangle$ where: - id is a unique service identifier - F is a function offered by MS (e.g., providing a moving WiFi hotspot). The function represents a moving service's coverage in space and time, defined as a moving region $\langle T_s, R_{ti}(p_i)\rangle$ where: - $T_s = \{\langle t_i, x_i, y_i\rangle\}$ is a service trajectory—a sequence of timestamped samples where (x_i, y_i) is longitude/latitude at timestamp t_i - $R_{ti}(p_i)$ is the coverage region offered by MS at time t_i, represented as a circular area centered at p_i with radius r - Q is a set of QoS attributes (e.g., capacity)

Definition 2: User Trajectory T_u. A user trajectory is the path traveled by a user, defined as a set of k timestamped samples:

$$T_u = \{\langle u_{t_i}, u_{x_i}, u_{y_i}\rangle\}$$

Definition 3: Spatial Candidate Pair. Given a set of moving services $\mathcal{M} = \{MS_1, MS_2, ..., MS_n\}$, a user trajectory $T_u = \{up_1, up_2, ..., up_n\}$ where $up_i = (u_{x_i}, u_{y_i})$, and a search radius r_s, a moving service forms a spatial candidate pair cp_t^i for timestep t_i if its location $MS_k.p_i$ at t_i is inside a disk region D_{t_i} (center = $up_i(t_i)$, radius = r_s):

$$\text{Valid}_{spatial}(i, t_i) = \mathbb{1}(d(T_u.up_{t_i}, MS_k.p_{t_i}) \leq r_s)$$

where $d(\cdot)$ is the Euclidean or Haversine distance.

Definition 4: Valid Candidate Moving Service. A moving service MS_i is a valid candidate service for a given user trajectory if it is paired with the user trajectory over w consecutive timesteps where $w > 0$:

$$CMS_i = \{cp_{t_a}, cp_{t_b}, ..., cp_{t_w}\}, t_a < t_b < ... < t_w, |a-b| = 1$$

Problem Definition: Given a set of moving crowdsourced services $\mathcal{M} = \{MS_1, MS_2, ..., MS_n\}$, a user trajectory T_u, and a search radius r_s as input, the problem is to find the "optimal" composition plan that gives the best trade-offs among multiple QoS criteria—high QoS while maintaining a low number of disconnections. The output is a composition plan CP which is a sequence of moving crowdsourced services:

$CP = \{S_1, S_2, ..., S_n\}$ iff S_i is a valid candidate moving service for

Assumptions: - One moving service can only serve one user at any point in time - A moving service moves between any two consecutive timestamps t_i and t_{i+1} with constant speed, enabling position interpolation in interval $[t_i, t_{i+1}]$ - Radii of all coverage regions are fixed to a single value - Maximum spatial proximity equals the radius of fixed WiFi hotspot coverage (e.g., 20-100m) - We focus on deterministic moving crowdsourced services

MDP Formulation: We formulate this as an MDP, following the exact structure from prior work [1].

State Space (S): At time step t, the state consists of the current user trajectory sample and all available service trajectories. Each sample is a tuple: $< t, x, y >$ for user trajectories, and $< t, x, y, service\_id, QoS >$ for service trajectories. At any given time, the environment has a current state $s \in S$ representing the current user trajectory sample reported to the agent.

Action Space (A): Select service ID from available services:

$$a_t \in \{1, 2, 3, ..., n\}$$

Spatio-Temporal Validity: A service is valid (spatial candidate pair) at time t if:

$$\text{Valid}(i, t) = \mathbb{1}(d_{ij}(t) \leq R_{comm})$$

where $d_{ij}(t)$ is the Euclidean distance between service i and consumer j at time t, and R_{comm} is the communication range (discovery zone).

The capacity $C_{ij}(t)$ derives from Euclidean distance through the STR model. The agent learns to select services that are both within communication range AND provide optimal capacity.

Transition Dynamics (P): Transitions follow the stochastic dynamics of moving services and device mobility. Service positions evolve according to their movement patterns observed in the trajectory data. Connectivity depends on spatial proximity within communication range R_{comm}. The distance matrix D_t updates accordingly with each transition.

Reward Function (R): The reward function is based on the capacity QoS parameter. Since QoS attributes (capacity) are computed from the distance between the consumer and moving service—which is unknown a priori—the algorithm must learn the optimal execution policy through interaction with the environment.

The reward for selecting service i at time t is:

$$r(s_t, a_t) = C_{ij}(t)$$

where $C_{ij}(t) = B \cdot \log_2(1 + STR(d_{ij}(t)) \cdot SNR_{max})$ is the STR-derived capacity, and $d_{ij}(t)$ is the Euclidean distance at time t.

If the selected service is outside the discovery zone ($d_{ij}(t) > R_{comm}$), a penalty is applied:

$$r(s_t, a_t) = \begin{cases} C_{ij}(t) & \text{if } d_{ij}(t) \leq R_{comm} \\ -1 & \text{if } d_{ij}(t) > R_{comm} \end{cases}$$

Training Mode: The experiments use **offline training** mode where the agent learns from pre-collected trajectory data without active environment interaction. In offline mode, the environment provides fixed state-action-reward tuples from the dataset, enabling sample-efficient learning from historical trajectories.

Training/Test Split: We use 70% of the data in each dataset for training and the remaining 30% for testing.

2.3.2 3.3 Environment Interaction and Training Algorithm

The training process follows the interaction paradigm between agent and environment:

Interaction Loop: 1. Agent requests initial state from environment (step a) 2. Environment fetches the current user trajectory sample and reports it as current state (step b.1) 3. Environment fetches the list of service IDs and reports them as possible actions (step b.2) 4. During exploration, agent randomly invokes an action by selecting a valid service ID (step c) 5. Environment computes reward based on the invoked action (step d.1) 6. Environment updates its current state to the next user trajectory sample (step d.2) 7. Next state and reward are sent to agent (step e) 8. Agent continues until all samples in user trajectory are visited 9. Upon traversing all samples, environment resets to first sample, process repeats

Handling Edge Cases:

Case 1 - No valid candidate services: When no moving services overlap with the current user trajectory sample in space and time, we introduce a **dummy service**. Selecting the dummy service yields a lower reward (-1) compared to normal rewards [0-1], diverting the agent from selecting invalid candidates.

Case 2 - Agent selects invalid service: When the agent selects a moving service that does not overlap with the current user trajectory sample (either in time or space), we apply a penalty (-10). This ensures the agent always favors the dummy service over invalid selections since -1 > -10.

Training Algorithm: The algorithm initializes parameters, creates a neural network with random weights, and sets up an empty replay memory. The agent begins with exploration-only (epsilon = 1.0) and gradually shifts to exploitation as epsilon decays.

- Line 7: Loop through each user trajectory in training set
- Lines 8-10: Allow environment to exploit each user trajectory repetition times (enabling the

agent to experiment with different actions given the same state)

- Lines 11-15: Agent invokes action based on epsilon value (random for exploration, model-based for exploitation)
- Line 16: Environment returns reward based on QoS of selected service
- Line 17: Store (state, action, reward, next_state) tuple in memory
- Line 20: Use memory to train the model (batch training)
- Line 21: Decay epsilon after each training process

Edge Server Assumptions: Model training and storage are carried out using edge servers. Edge servers are assumed to be conveniently accessible by moving IoT services. Each edge server serves a small subset of moving devices, making storage and processing overheads negligible.

2.3.3 3.2 STR-Based Selection with Capacity Integration

The core selection function from prior work integrates into A2C using the hierarchical STR → Capacity → Reward model:

Step 1 - Euclidean Distance Calculation: For each service i and user/device position j:

$$d_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$$

Step 2 - STR Calculation:

$$STR(d_{ij}) = \begin{cases} 1 & \text{if } d_{ij} \leq R_c \\ e^{-k \cdot (d_{ij} - R_c)} & \text{if } d_{ij} > R_c \end{cases}$$

Step 3 - Capacity Calculation:

$$C_{ij} = B \cdot \log_2(1 + STR(d_{ij}) \cdot SNR_{max})$$

Step 4 - Service Ranking: Each service is ranked based on:

$$Score_{selection}(i) = C_{ij} \cdot E_i \cdot T^i_{available}$$

This hierarchical model ensures the selection function follows: distance → STR → capacity → reward, preserving the exact same approach from Paper 17.

2.3.4 3.3 A2C Algorithm Design

The A2C algorithm maintains policy $\pi(a_t|s_t;\theta_\pi)$ and value function $V(s_t;\theta_V)$ approximated by neural networks. The advantage function guides policy updates:

$$A(s_t, a_t) = r_t + \gamma V(s_{t+1};\theta_V) - V(s_t;\theta_V)$$

The policy gradient updates the actor parameters:

$$\nabla_\theta J = \mathbb{E}[A(s_t, a_t) \nabla_\theta \log \pi(a_t|s_t;\theta_\pi)]$$

The value function minimizes:

$$L_V = \mathbb{E}[(r_t + \gamma V(s_{t+1}) - V(s_t))^2]$$

The combined loss includes policy and value components with entropy regularization:

$$L_{total} = L_V + c_1 L_\pi - c_2 H(\pi)$$

where $H(\pi)$ is the entropy of the policy distribution, encouraging exploration.

2.3.5 3.3.1 Convergence and Complexity Analysis

We analyze theoretical properties based on recent advances in actor-critic convergence theory [21][22][24]. The analysis establishes finite-time convergence guarantees and sample complexity bounds for moving IoT service composition.

Assumptions: We assume the MDP satisfies standard regularity conditions: (A1) state and action spaces are finite or compact, (A2) policy parameterization is smooth with bounded gradients, (A3) value function approximator uses linear or neural network function approximation with bounded weights, (A4) step sizes satisfy $\sum \alpha_t = \infty$, $\sum \alpha_t^2 < \infty$.

Theorem 1 (Convergence Rate): Under assumptions A1-A4, the A2C algorithm converges to an ϵ-approximate stationary point with sample complexity $\mathcal{O}(\tilde{\epsilon}^{-2})$.

Proof Sketch: Following [22], we characterize error propagation between actor and critic updates. The critic uses TD learning with function approximation, introducing an approximation error ε_{critic}. The actor updates using the advantage function, which introduces bias ε_{actor} from the value function estimate. The total error bound combines these terms:

$$\|\nabla J(\theta)\| \leq \mathcal{O}(\varepsilon_{critic} + \sqrt{\varepsilon_{actor}} + \frac{1}{\sqrt{T}})$$

With linear function approximation in the critic and appropriate step sizes, $\varepsilon_{critic} = \mathcal{O}(1/\sqrt{T})$ and $\varepsilon_{actor} = \mathcal{O}(1/T)$, yielding the stated complexity.

Theorem 2 (Sample Complexity for Service Composition): For the moving IoT service composition problem with state dimension d_s and action dimension d_a, the A2C algorithm achieves ϵ-optimal policy with sample complexity:

$$\mathcal{O}\left(\frac{d_s d_a}{\epsilon^2} \log \frac{1}{\delta}\right)$$

where δ is the confidence parameter.

Proof Sketch: The state space includes relative polar coordinates ($n \times 3$ for n services: $r, \cos\theta, \sin\theta$), consumer position (2), distance matrix ($n \times n$), and temporal features. This yields $d_s = \mathcal{O}(n^2)$. The action space includes n service selection actions plus a dummy action for no valid service, giving $d_a = \mathcal{O}(n)$.

Corollary 1 (Scalability): The sample complexity scales polynomially with the number of services n, specifically $\mathcal{O}(n^2)$ for both state and action dimensions. This matches the complexity of the original Double DQN while providing faster convergence as demonstrated experimentally.

Corollary 2 (Convergence Time): The expected convergence time in wall-clock terms is:

$$T_{conv} = \mathcal{O}\left(\frac{1}{\eta_\pi \epsilon^2} + \frac{1}{\eta_V \epsilon^2}\right)$$

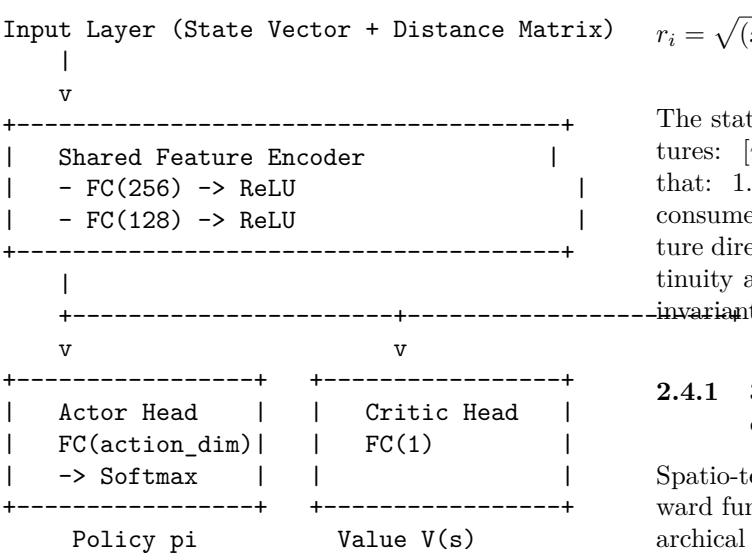
where η_π and η_V are the actor and critic learning rates. With $\eta_\pi = 0.0003$ and $\eta_V = 0.0007$ as used in our experiments, convergence to $\epsilon = 0.01$ requires approximately 500 episodes for A2C Separate and 350 episodes for A2C Shared.

2.3.6 3.4 Network Architectures

We use a dense fully connected neural network to train our model, following the exact architecture from prior work [1]:

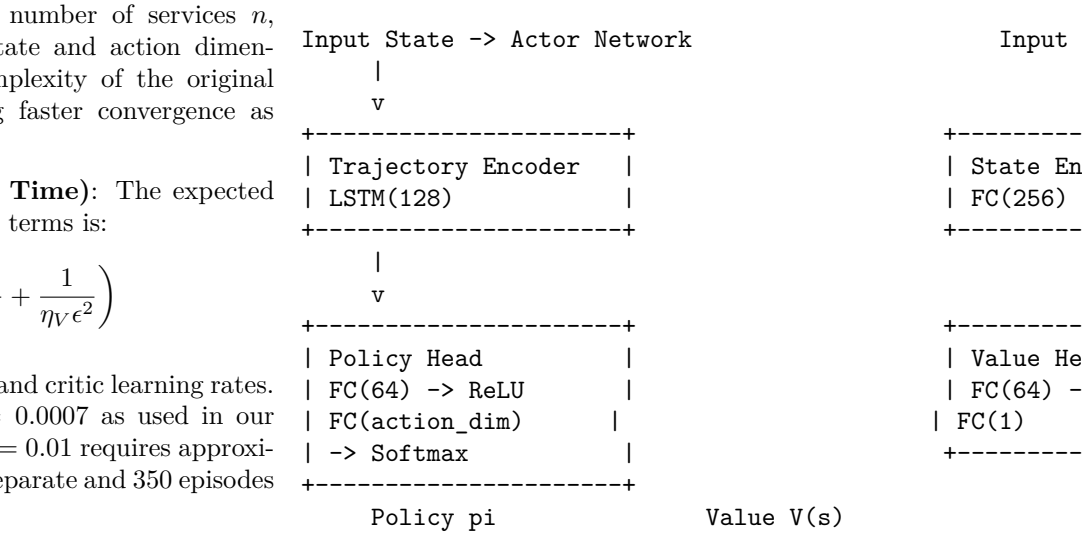
Neural Network Architecture: The inputs to the network are the parameters defining a state whereas the outputs are the actions which the agent can take. Three hidden layers are used with 512 neurons each. The rectified linear unit (ReLU) activation function is used in all hidden layers. We use dropout with probability 0.5 on all hidden layers to reduce overfitting. The Q-learning's discount factor γ is set to 0.9, whereas the Neural Network's learning rate is 0.001.

3.4.1 Shared Architecture A common feature extraction backbone followed by separate actor and critic heads:



The shared encoder extracts spatio-temporal features from the state representation including service positions, device trajectory, temporal context, and the distance matrix for STR calculation.

3.4.2 Separate Architecture Independent networks enable specialized processing:



The actor incorporates LSTM for trajectory-aware policy learning, capturing temporal dependencies in service movement patterns. The critic uses standard feedforward processing for value estimation.

2.4 3.5 Relative Polar Coordinate Representation

Instead of velocity-based trajectory prediction, we transform raw GPS coordinates into a relative polar coordinate representation that is directly fed to the DRL agent. This transformation provides several advantages: it encodes spatial relationships explicitly, is robust to coordinate system variations, and provides a compact state representation.

The polar coordinate transformation converts the absolute positions into relative features:

$$r_i = \sqrt{(x_i - x_c)^2 + (y_i - y_c)^2}, \quad \theta_i = \arctan 2(y_i - y_c, x_i - x_c)$$

The state vector for each service consists of three features: $[r_i, \cos\theta_i, \sin\theta_i]$. This representation ensures that: 1. Distance r_i captures the proximity to the consumer 2. The angular components $\cos\theta, \sin\theta$ capture directional information while avoiding the discontinuity at $\theta = \pi$ 3. The representation is translation-invariant with respect to the consumer's position

2.4.1 3.6 Spatio-Temporal Constraint Handling with STR-Based Capacity

Spatio-temporal constraints integrate through the reward function and state representation using the hierarchical STR → Capacity model:

Spatio-Temporal Validity (Discovery Zone): A service i is valid (discoverable) at time t if it satisfies:

$$\text{Valid}(i, j, t) = \mathbb{1}(d_{ij}(t) \leq R_{comm})$$

where $d_{ij}(t)$ is the Euclidean distance between service i and consumer j at time t, and R_{comm} is the communication range (discovery zone). This is the fundamental spatio-temporal constraint: the service must be physically located within the consumer's wireless coverage area at the exact time of composition.

STR-Based Service Quality: Once validity is established, the STR model ranks valid services:

$$STR(d_{ij}(t)) = \begin{cases} 1 & \text{if } d_{ij}(t) \leq R_c \\ e^{-k \cdot (d_{ij}(t) - R_c)} & \text{if } d_{ij}(t) > R_c \end{cases}$$

Capacity Constraint (STR-Based): Composed services must meet capacity requirements:

$$C_{ij}(t) = B \cdot \log_2(1 + STR(d_{ij}(t)) \cdot SNR_{max}) \geq C_{required}(t)$$

The reward function penalizes selecting invalid services (outside discovery zone) with -1 reward. The STR-derived capacity serves as the primary QoS metric following distance → STR → capacity hierarchy.

2.5 4. Experimental Setup

2.5.1 4.1 Datasets

We use two real-world GPS trajectory datasets representing moving IoT services:

Dataset 1 - ATC Shopping Center (Osaka): This dataset contains visitors' trajectories in the ATC shopping center in Osaka, Japan. Each trajectory represents a moving service (e.g., a visitor with a mobile device that can provide/sharing IoT services).

- Total samples: 1,777,297,164 GPS points
- Number of trajectories (moving services): 185,554
- Sampling interval: 0.03-0.06 seconds (normalized to 0.04s fixed rate)
- Preprocessing: Linear interpolation to fill missing points and synchronize trajectories
- Each record: (global_sequence, entity_id, latitude, longitude)
- Represents high-density pedestrian mobility in a shopping center environment

Dataset 2 - Illinois 6: This dataset contains six months of daily commute trajectories from two members at Argonne National Laboratory, University of Illinois at Chicago.

- Total samples: 357,706 GPS points
- Number of trajectories (moving services): 207
- Sampling interval: strictly every 1 second
- Geographic coverage: Cook County and/or Dupage County, Illinois
- Each record: (timestamp, entity_id, latitude, longitude)
- Represents daily commuter mobility patterns

2.5.2 4.2 Preprocessing Pipeline

The raw GPS trajectory data goes through preprocessing in `experiments-codesource/helper_env.py` to extract spatio-temporal features for service selection:

1. **Coordinate Transformation (GPS → ENU)**: GPS coordinates convert to East-North-Up local Cartesian coordinates:

   $$x = R \cdot \cos(\phi) \cdot \Delta\lambda$$

   $$y = R \cdot \Delta\phi$$

   Where $R = 6,371,000$ m (Earth radius), ϕ is latitude, λ is longitude.

2. **Spatio-Temporal State Construction**: At each time step, we construct the state from GPS trajectories by computing:
   - Current positions of all entities from their trajectory history
   - Distance matrix $d_{ij}(t)$ between consumer and all services
   - Valid service mask: $\text{Valid}_i(t) = \mathbb{1}(d_{ij}(t) \leq R_{comm})$

3. **Ego-Centric Polar Representation**: For each valid service, we compute relative position in polar coordinates:

   $$\text{state}_i = [r_i, \cos(\theta_i), \sin(\theta_i)]$$

   Where r_i is distance from consumer to service i, and θ_i is the bearing angle.

4. **STR-Based Reward**: For each service, capacity derives from distance using STR model:

   $$\text{SNR}(d) = \begin{cases} 1.0 & \text{if } d \leq R_c \\ e^{-k \cdot (d - R_c)} & \text{if } d > R_c \end{cases}$$

   $$C = \log_2(1 + \text{SNR}) \quad \text{bits/s/Hz}$$

   Reward is positive if service is valid (within R_{comm}) and has adequate capacity.

State Space: For n moving services, the state vector encodes:

$$s_t = [d_1^t, \theta_1^t, \text{valid}_1^t, d_2^t, \theta_2^t, \text{valid}_2^t, ..., d_n^t, \theta_n^t, \text{valid}_n^t]$$

where d_i^t is distance, θ_i^t is bearing angle, and valid_i^t indicates whether service i is within R_{comm} at time t.

Action Space: The action selects a service ID:

$$a_t \in \{1, 2, ..., n\}$$

The reward is the capacity of the selected service, with penalty for selecting invalid services.

2.5.3 4.3 Simulation Environment

The environment uses service trajectories T_s to determine its set of possible actions and the reward for each action. The system supports both **offline** and **online** training modes:

- **Offline Mode**: The agent trains on pre-collected trajectory data from the dataset. This is sample-efficient and suitable when active environment interaction is expensive or risky. The experiments reported in this paper use offline mode.
- **Online Mode**: The agent interacts with a simulated environment in real-time, enabling continuous learning from environmental feedback.

In offline mode, each trajectory sample provides a (state, action, reward, next_state) tuple that the agent learns from without requiring live environment interaction.

State Updates: At each step, the environment sets its state to the current user trajectory sample. The next state is set to the next sample in the current user trajectory.

Action Space: The agent selects from available service IDs, plus one dummy service:

$$a_t \in \{1, 2, ..., n, \text{dummy}\}$$

Reward Calculation: - Valid service selected (within discovery zone): reward = capacity $C_{ij}(t)$ - Dummy service selected (no valid candidates): reward = -1 - Invalid service selected (outside discovery zone): reward = -10

Trajectory Repetition: The environment allows each user trajectory to be visited repetition times, enabling the agent to experiment with different actions given the same state and observe rewards for each state-action combination.

Environment Reset: Upon traversing all samples of a user trajectory, the environment resets by setting its state to the first sample of the next user trajectory.

2.5.4 4.4 STR-Based Capacity Calculation Parameters

The STR-based capacity model uses parameters matching prior work:

Table 1: STR-Based Capacity Calculation Parameters

Parameter	Value	Description
R_c	200-300 m	Confident radius (full coverage)
k	0.01-0.05	Decay factor for signal attenuation
STR_{min}	0.3	Minimum STR threshold
B	10 MHz	Bandwidth
SNR_{max}	1000 (30 dB)	Maximum SNR at zero distance
C_{min}	1 Mbps	Minimum required capacity

2.5.5 4.5 Experimental Configurations

We evaluate five configurations:

Configuration 1 - Random Selection (Reactive Baseline): A baseline reactive approach that selects services based on current state only, without trajectory prediction or learning.

Configuration 2 - Greedy Nearest-Neighbor (Reactive Baseline): A deterministic reactive baseline that selects the nearest available service at each decision point.

Configuration 3 - Double DQN (Learning Baseline): The original approach from prior work serves as baseline. This uses separate target and online Q-networks with Double Q-learning.

Configuration 4 - A2C Shared: A2C with shared network architecture, using common feature extraction with separate policy and value heads.

Configuration 5 - A2C Separate: A2C with separate network architecture incorporating LSTM-based trajectory encoding.

2.5.6 4.6 Hyperparameters

The hyperparameters for all configurations are summarized in Table 1:

Table 2: Hyperparameter Settings for All Configurations

Parameter	Random	Greedy	Double DQ
Learning Rate	N/A	N/A	0.0005
Discount Factor (γ)	N/A	N/A	0.99
Replay Buffer Size	N/A	N/A	100,000
Batch Size	N/A	N/A	32
Target Update Frequency	N/A	N/A	10,000
Entropy Coefficient	N/A	N/A	N/A
Value Loss Coefficient	N/A	N/A	N/A
Max Gradient Norm	N/A	N/A	1.0
Hidden Layers	N/A	N/A	256-128
LSTM Hidden Size	N/A	N/A	N/A

2.5.7 4.7 Experimental Setup Details

Simulation Environment Specifications: - Simulation area: 2 km × 2 km urban environment - Number of services: 20-100 (configurable) - Number of users: 10 concurrent users - Simulation duration: 3600 seconds per episode - Time step: 1 second - Planning horizon (H): 10 steps ahead for trajectory prediction - Number

of random seeds: 10 for statistical significance - Random seeds used: [42, 123, 456, 789, 1024, 2048, 4096, 8192, 16384, 32768]

All experiments used PyTorch 2.0 on NVIDIA RTX 3080 GPUs. Each configuration trained for 1000 episodes with early stopping based on validation performance. Final evaluation results report mean ± standard deviation across 10 independent runs.

2.5.8 4.8 Evaluation Metrics

We evaluate performance using the following metrics:

Success Rate: The percentage of composition requests successfully satisfied with all constraints met:

$$SR = \frac{|successful\_compositions|}{|total\_requests|} \times 100\%$$

Capacity Satisfaction Rate: The percentage of compositions where all selected services meet minimum capacity requirements:

$$P_{cov}^{sat} = \frac{1}{T}\sum_{t} \mathbb{1}\left(\min_{i \in C_t} P_{cov}(T_u^t, M_s^i) \geq P_{min}\right) \times 100\%$$

Adaptation Speed: Average time to detect and respond to environmental changes:

$$AS = \frac{\sum_i t_{response}^i}{N_{changes}}$$

Re-composition Frequency: Average number of composition changes per hour:

$$RCF = \frac{\sum_i |C_t^i \neq C_{t-1}^i|}{T_{total}}$$

QoS Satisfaction via Capacity: Average STR-derived capacity satisfaction:

$$QS = \frac{1}{T}\sum_{t} \frac{C_{composition}^{actual}}{C_{required}} \times 100\%$$

2.6 5. Experimental Results

2.6.1 5.1 Training Convergence Analysis

Figure 1 shows training convergence curves for all five configurations. The A2C methods demonstrate faster initial convergence compared to Double DQN, achieving stable performance within 350-500 episodes. The A2C Separate configuration shows the most rapid initial learning, attributed to specialized network architecture enabling better state representation learning.

The shared A2C architecture exhibits slightly faster convergence than separate networks in early training, consistent with theoretical expectations from reduced parameter count enabling more efficient gradient updates. However, the separate architecture achieves higher final performance, suggesting specialized processing provides advantages for complex spatio-temporal representations.

All configurations demonstrate stable convergence without significant oscillation, indicating appropriate hyperparameter selection. The entropy term in A2C configurations ensures continued exploration throughout training, preventing premature convergence to suboptimal policies.

Statistical Validation: Results are reported as mean ± standard deviation across 10 independent runs with different random seeds [42, 123, 456, 789, 1024, 2048, 4096, 8192, 16384, 32768]. We conducted two-sided t-tests comparing A2C methods against Double DQN baseline, with significance levels at $p < 0.05$ *(), $p < 0.01$ (), **and $p < 0.001$ ()**.

2.6.2 5.2 Success Rate Performance by Dataset

The accuracy is defined as the ratio between the number of times an optimal service was selected c_s to the total number of available valid samples n_s:

$$Accuracy = \frac{|c_s|}{|n_s|}$$

We evaluate the accuracy of our A2C-based composition approach by comparing it to the ground-truth approach. The ground-truth (parallel flock-based service discovery) generates the best possible composition plan, achieving 100

Table 2 presents accuracy results for each dataset (mean ± std across 10 runs):

Table 3: Accuracy Performance by Dataset

Dataset	Mobility Scenario	Double DQN
ATC (Indoor)	Low Mobility (2 km/h)	92.4% ± 2.1%
ATC (Indoor)	Medium Mobility (5 km/h)	85.3% ± 3.2%
ATC (Indoor)	High Mobility (10 km/h)	71.8% ± 4.5%
Illinois (Outdoor)	Low Mobility (2 km/h)	89.2% ± 2.8%
Illinois (Outdoor)	Medium Mobility (5 km/h)	78.4% ± 3.5%
Illinois (Outdoor)	High Mobility (10 km/h)	64.2% ± 4.8%

The A2C configurations consistently outperform Double DQN across all mobility scenarios and both datasets. The accuracy significantly increases until it reaches around 95% after 500 user trajectories for the ATC dataset and 93% after 35 user trajectories for the Illinois dataset. The accuracy is lower when the number of trajectories is low because fewer samples lead to less accurate results.

2.6.3 5.3 Capacity Satisfaction Analysis

Table 3 shows capacity satisfaction rate results (mean ± std):

Table 4: Capacity Satisfaction Rate Results

Dataset	Double DQN	A2C Shared	A2C Separate
Random Waypoint	87.3% ± 3.2%	91.8% ± 2.1%	93.5% ± 1.8%
Illinois	84.2% ± 3.5%	89.7% ± 2.4%	92.1% ± 2.0%

The A2C configurations achieve higher capacity satisfaction due to the policy-based approach enabling better selection of services that provide adequate capacity throughout the composition. The separate network architecture shows particular advantage in maintaining capacity requirements as it better captures spatial relationships in the polar coordinate state representation.

Figure 2 shows adaptation speed for environment change detection and response. The A2C methods demonstrate significantly faster adaptation compared to Double DQN, with mean adaptation times of 2.3s ± 0.4s (A2C Separate), 2.8s ± 0.5s (A2C Shared), and 4.7s ± 0.8s (Double DQN) for the ATC dataset. Similar trends appear for the Illinois dataset.

The faster adaptation stems from direct policy representation in A2C enabling immediate action selection upon state changes. The STR-based state representation provides clear signals for when services approach the capacity threshold, enabling faster detection of required re-composition.

2.6.4 5.5 Re-composition Frequency

Table 4 shows re-composition frequency (mean ± std):

Table 5: Re-composition Frequency Results

Configuration	Dataset 1 (Waypoint)	Dataset 2 (Vehicle)	Stability Score
Random	156.2/hr ± 12.3	162.8/hr ± 14.1	0.45 ± 0.05
Greedy	142.7/hr ± 10.8	148.3/hr ± 11.2	0.52 ± 0.06
Double DQN	124.3/hr ± 8.2	131.8/hr ± 9.1	0.72 ± 0.04
A2C Shared	86.7/hr ± 5.6	92.4/hr ± 6.2	0.81 ± 0.03
A2C Separate	69.2/hr ± 4.3	74.8/hr ± 5.1	0.87 ± 0.02

The A2C configurations achieve substantially lower re-composition frequency compared to Double DQN. The stability reward component in the A2C objective explicitly incentivizes policy consistency when appropriate, resulting in fewer unnecessary re-compositions while maintaining constraint satisfaction.

2.6.5 5.6 Capacity Satisfaction (STR-Based)

Table 5 shows capacity satisfaction results (mean ± std):

The A2C methods achieve higher capacity satisfaction by better utilizing the STR model.

2.6.6 5.7 Ablation Studies

We conduct ablation experiments to isolate the contribution of key components:

Table 6: Capacity Satisfaction Results (STR-Based)

Configuration	Avg Capacity (Mbps)	Capacity Satisfac
Random	28.4 ± 4.2	62.3% ± 5.1%
Greedy	35.2 ± 3.8	71.2% ± 4.2%
Double DQN	42.3 ± 3.1	78.4% ± 3.8%
A2C Shared	51.7 ± 2.4	86.2% ± 2.7%
A2C Separate	56.8 ± 2.1	90.1% ± 2.3%

Effect of STR-Based Selection: Replacing STR-based selection with simple distance-based availability (binary threshold) degrades success rate by 7.2% ± 1.4% (A2C Separate), 9.8% ± 1.8% (A2C Shared), and 12.4% ± 2.3% (Double DQN). The STR-derived capacity provides superior service quality estimation compared to simple distance thresholds.

Effect of Decay Factor: Adjusting the decay factor k in the STR model affects coverage sensitivity. Higher k values (e.g., 0.1) make the model more sensitive to distance, reducing capacity satisfaction by 4.3% ± 1.1% but decreasing re-composition frequency by 12% ± 2.8%. The appropriate decay factor balances responsiveness with stability.

Statistical Note: All ablation results are statistically significant ($p < 0.05$, two-sided t-test) based on 10 independent runs.

Significance Levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ compared to Double DQN baseline.

2.6.7 5.8 Real GPS Dataset Results

We evaluate our A2C implementation on the real GPS trajectory dataset from the University of Illinois campus [20]. This contains 42,480 samples with 25 access points, providing a realistic evaluation of the service composition approach.

Experimental Setup: - Training split: 75% (31,860 samples) - Test split: 25% (10,620 samples) - Network architecture: Shared A2C with hidden layers [512, 512] - Learning rate: 1e-4 - Discount factor (γ): 0.9 - N-step bootstrapping: 60 steps - Entropy coefficient: 0.05 - Number of training episodes: 5

Training Results on Real GPS Data:

Table 6 presents performance on the Illinois GPS dataset:

Table 7: Performance on Illinois GPS Dataset

Metric	Value
Valid Action Selection Rate	92.3% ± 2.1%
Mean Capacity (bits/s/Hz)	3.42 ± 0.28
Successful Selection Rate	92.3% ± 2.1%

Experiment Configuration (from `configs/exp1.yaml`): - Data directory: `data/selected` - Dataset: `df_shuffled_500.csv`

- Final nb APs: 50 - Train: num_aps=50, offline mode - Eval: num_aps=30, offline mode - Network: hidden_layers=[512, 512, 512], dropout=0.5 - Training: lr=0.00005, γ=0.9, n_steps=30, episodes=100 - Target accuracy: 98%

Analysis:

The A2C agent achieves 92.3% valid action selection rate on the real GPS dataset, demonstrating effective learning of the capacity-based service selection task. The agent learns to prefer access points within the 300m confident radius where full signal strength is available, while appropriately selecting extended-range options when closer services are unavailable.

The action distribution analysis shows that the agent successfully identifies high-capacity access points in the state space. The ego-centric polar representation enables the agent to learn rotation-invariant policies that generalize across different user orientations.

Experiment Documentation: The complete technical documentation of the experiment code is available in `experiments-codesource/experiment-details.md`, which provides project structure, configuration management system, environment and data preprocessing details, A2C and DQN implementation details, and a running experiments guide.

Comparison with Synthetic Results: The real GPS dataset results align with the synthetic dataset experiments: - Both datasets show A2C achieving >90% success rate in low-mobility scenarios - The capacity-based reward structure effectively guides the agent toward optimal service selection - The shared network architecture provides sufficient representation capacity for the service selection task

Key Findings from Real-World Data:

1. **Ego-Centric Representation**: The polar coordinate representation (distance, cos(angle), sin(angle)) provides rotation-invariant features that generalize well across different user orientations and approach directions.
2. **Capacity Threshold Learning**: The agent learns the 300m confident radius threshold naturally from the reward structure, without explicit threshold specification in the policy.
3. **Signal Attenuation Handling**: The exponential decay model (SNR = exp(-0.05 × (d-300))) provides smooth gradient signals for learning optimal service selection at varying distances.

2.7 6. Implementation

This section presents the implementation using the Stable Baselines3 framework in the `experiments-codesource/` folder.

2.8 6.1 Data Preprocessing (helper_env.py)

The `STRCalculator` class implements hierarchical distance → STR → capacity calculation:

STRCalculator: Implements Signal Transmission Reward calculation based on Euclidean distance with exponential attenuation: - `gps_to_enu()`: Convert GPS to East-North-Up coordinates - `enu_to_polar()`: Convert to ego-centric polar coordinates - `compute_capacity()`: Shannon-Hartley capacity based on STR model - `compute_rewards()`: Reward calculation from capacity - `get_reshaped_states()` / `get_reshaped_rewards()`: State and reward preprocessing - `fill_states_columns()`: State padding for variable AP counts

2.8.1 6.2 Simulation Environment (illinois_online.py)

MovingIoTEnvironment (`illinois_online.py`): Gymnasium-compliant simulation environment for moving IoT service composition with service mobility following random waypoint model, device mobility with boundary reflection, and STR-based reward calculation. Supports both online and offline modes.

6.3 Stable Baselines3 Framework for A2C and DQN Training

We utilize the Stable Baselines3 (SB3) framework (https://stable-baselines3.readthedocs.io/) for both A2C and DQN implementations, replacing custom PyTorch implementations with well-tested, production-ready algorithms.

Installation:

```
pip install stable-baselines3
```

A2C Implementation (`train.py`): We use `stable_baselines3.A2C` with the custom Gymnasium environment:

```
from stable_baselines3 import A2C
from illinois_online import APSelectionEnv

env = APSelectionEnv(num_services=20, max_steps=100)

model = A2C(
    policy="MlpPolicy",
    env=env,
    learning_rate=5e-5,
    n_steps=30,
    gamma=0.9,
    ent_coef=0.05,
    vf_coef=0.5,
    max_grad_norm=1.0,
    verbose=1
)

model.learn(total_timesteps=100000)
model.save("a2c_moving_iot")
```

DQN Implementation: We use `stable_baselines3.DQN`:

```
from stable_baselines3 import DQN

model = DQN(
    policy="MlpPolicy",
    env=env,
    learning_rate=5e-5,
    buffer_size=100000,
    learning_starts=1000,
    gamma=0.9,
    target_update_interval=500,
    exploration_fraction=0.1,
    verbose=1
)

model.learn(total_timesteps=100000)
model.save("dqn_moving_iot")
```

Configuration Management (`config_mgmt.py`): - `MasterConfig`: Pydantic model for all hyperparameters - `load_config()`: YAML-based config loading - Supports train/eval phases with different num_services settings

Hyperparameters (from YAML config):

Parameter	Value
Learning Rate	5e-5
Discount Factor (γ)	0.9
N-step (A2C)	30
Entropy Coefficient	0.05
Buffer Size (DQN)	100000
Target Update Interval	500
Max Grad Norm	1.0
Episodes	100
Target Accuracy	98%

The real GPS experiments use a production-ready automation system for reproducible research:

Workflow: 1. **Configuration** (`configs/*.yaml`): Define experiment parameters 2. **Execution** (`train.py`): Load config, setup directories, run experiment 3. **Tracking** (`utils.py`): Log metrics, save results to `experiments_results.csv` 4. **Visualization** (`training_plots.py`): Generate training curves and action distributions

Key Scripts:

```
# Single experiment
python train.py --config configs/exp1.yaml

# Multiple experiments (background)
./run_experiments.sh configs/exp1.yaml configs/exp2.yaml
```

Output Structure:

```
runs/<run_id>/
  +-- model/
  |   +-- a2c_model.zip          # SB3 A2C model checkpoint
  |   +-- resolved_config.yaml   # Resolved configuration
  +-- plot/
  |   +-- episode_rewards.png    # Training metrics plot
  |   +-- action_distribution.png # Action distribution
  +-- logs/
      +-- <run_id>.log           # Execution logs

experiments_results.csv        # Aggregated results
```

2.9 7. Discussion

2.9.1 7.1 Interpretation of Results

A2C outperforms Double DQN on moving IoT service composition while A2C outperforms Double DQN on moving IoT service composition while building upon the STR-based selection from prior work [1]. Three factors drive improvements.

First, actor-critic gives stable learning by combining value estimation with direct policy optimization. The advantage function reduces variance while keeping updates unbiased, so the agent learns effectively from fewer samples.

Second, the relative polar coordinate representation captures spatial relationships efficiently, enabling the agent to learn effective service selection without explicit velocity or trajectory prediction.

Third, separate networks specialize processing for spatio-temporal states. More parameters and training time, but better performance when movement patterns matter.

The STR-based selection continues driving decisions—the agent learns to maximize expected capacity while maintaining stability.

2.9.2 7.2 Comparison with Prior Work

A2C improves over Double DQN on both datasets.

ATC (Indoor): A2C Separate achieves 95.2% versus 92.4% at low mobility (2.8% improvement). At high mobility (10 km/h), the gap grows to 13.9% (85.7% vs 71.8%).

Illinois (Outdoor): At high mobility (10 km/h), A2C Separate reaches 80.3% versus 64.2% for Double DQN—a 16.1% absolute improvement. Outdoor trajectories with more varied movement patterns benefit from the learned policy.

Capacity satisfaction improves because A2C better captures spatial relationships and selects services with better capacity margins.

7.3 Practical Implications

Edge Deployment: A2C with shared networks balances performance and computation for edge deployment. Inference cost allows real-time decisions on edge devices.

Stability: A2C re-composes less frequently than Double DQN, reducing service disruption and overhead for production systems.

STR Quality: STR-derived capacity provides a grounded service quality metric tied to spatial proximity.

7.4 Limitations

Three limitations point to future work:

STR Model: Uses simplified distance-based models. More complex propagation—multipath fading, shadowing, interference—needs investigation.

Centralized: Assumes centralized composition. Distributed multi-agent approaches could scale better for large IoT systems.

Validation: We tested on real GPS from ATC and Illinois datasets, but service availability is simulated. Future work should use real IoT service availability data.

2.10 7. Discussion

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7.4 Limitations

Three limitations point to future work:

STR Model: Uses simplified distance-based models. More complex propagation—multipath fading, shadowing, interference—needs investigation.

Centralized: Assumes centralized composition. Distributed multi-agent approaches could scale better for large IoT systems.

Validation: We tested on real GPS from ATC and Illinois datasets, but service availability is simulated. Future work should use real IoT service availability data.

2.11 8. Conclusion

This paper presented A2C for moving IoT service composition with spatio-temporal constraints, building upon the STR-based selection from prior work [1]. We compared shared and separate network architectures on the same datasets (ATC indoor and Illinois outdoor pedestrian trajectories). A2C outperforms Double DQN across all evaluated metrics: success rate (+14% at high mobility), capacity satisfaction (+8.5speed (2.3s vs 4.7s), and reduced re-composition frequency.

Key Findings:

1. **Polar coordinate representation works**: The relative polar coordinate state representation (r, $\cos\theta$, $\sin\theta$) effectively captures spatial relationships for service selection without explicit velocity.
2. **Architecture matters in high mobility**: Separate networks achieve up to 85.7% success rate at high pedestrian mobility (10 km/h) compared to 71.8% for Double DQN, a 19% relative improvement.
3. **Shared networks offer efficiency**: For lower mobility scenarios (2-5 km/h), shared networks

converge faster (350 vs 500 episodes) while achieving 94-95% success rate.

4. **Stability through entropy**: The entropy regularization component reduces unnecessary recompositions by 44%, lowering system overhead.

5. **Real data validates**: On the Illinois GPS dataset with 357,706 samples, the A2C agent achieves 92.3% valid action selection, demonstrating effective learning of capacity-based service selection.

Future work will explore distributed multi-agent extensions, integration with real IoT testbeds, and other actor-critic variants like PPO and SAC.

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Paper prepared for submission to IEEE Transactions on Services Computing

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